DOHYUN KANG

Embedded Software Engineer | Automotive Communications | Test Automation

Incheon, South Korea | | | github.com/DohyunKang

SUMMARY

Engineer with hands-on experience in automotive validation, CAN/CAN FD/UDS communication, measurement automation, and embedded prototyping. Built verification and control tools in C#, Python, and C/C++ and delivered smart parking, safety IoT, and RF/antenna projects.

EDUCATION

Incheon National University *Mar 2019 – Feb 2025*

B.S. in Information & Communication Engineering | GPA: 3.93/4.5 overall, 4.05/4.5 major *Incheon, South Korea*

EXPERIENCE

TechWays — Embedded/Test Automation Intern *Aug 2024 – Feb 2025*

- Built C# and LabVIEW tools for ECU/power-equipment control, serial communication, PWM handling, and real-time data visualization.
- Improved multi-frame UDS reception with Flow Control logic, queue-based logging, and timed batch processing.

ALPS ALPINE Korea — Automotive Validation Intern *Jan 2024 – Feb 2024*

- Verified vehicle switch products in CAN/CAN FD environments using CANoe, CANalyzer, and VT System under lab and vehicle test conditions.
- Analyzed signal behavior and validation criteria while working with TR/RS-aligned outputs and issue tracking.

High-Frequency Antenna Design Lab — Undergraduate Researcher *Sep 2022 – Dec 2023*

- Designed antennas in CST and evaluated S11, radiation patterns, and 28 GHz propagation using VNA and spectrum-analysis equipment.
- Contributed to an ESP32-based beehive monitoring system with DHT/MQ sensors and Wi-Fi data transmission.

SELECTED PROJECTS

Smart Parking System for Villas with Small Parking Spaces *2024*

- Built an AI-assisted parking system with live camera monitoring, license-plate recognition, occupancy tracking, and departure-time prediction.
- Used OpenCV, Tesseract OCR, Django/Node.js, MySQL, and HTML/CSS/JavaScript.

Safety Helmet with Communication Function for Accident Prevention *2023*

- Developed a smart helmet prototype using SAMD21G, gyroscope, air-quality sensing, buzzer alerting, and LoRa-based communication.

TECHNICAL SKILLS & AWARDS

Languages/Tools: C, C++, C#, Python, CANoe, CANalyzer, NI DAQ, LabVIEW, WinForms, Django, MySQL, OpenCV, Tesseract OCR

Domains: CAN, CAN FD, UDS, Ethernet, embedded systems, ECU verification, serial communication

Training: Vehicle CAN/CAN-FD, Vehicle Ethernet (Sungkyunkwan University, 2024)

Awards:

- Industry-Academia Capstone Design Silver Prize, Incheon National University (2023)
- 2024 Hanium ICT Mentoring Contest Award, Korea Information Industry Association (2024)
- ACK 2024 Paper Award, Korea Information Processing Association (2024)